
Diabetes Prediction and Transformers: a Tabular Approach

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Abstract

This project explores transformer-based architectures for early diabetes prediction using physiological and demographic tabular data (aka blood glucose level, sex, etc.). We propose two parallel approaches: (i) a custom transformer architecture specifically designed for tabular data that leverages learnable embeddings and multi-head self-attention layers, and (ii) a multi-layer Deep Neural Network, reigning as our baseline for this model to compare against the transformer and note whether transformer architectures provide an advantage in diabetes prediction scenarios. Thus, Our goal with the transformers is to capture complex inter-feature interactions and improve upon classical deep neural network baselines. In this report, we describe our data preparation, including methods aimed to mitigate the inherent class imbalance of the dataset, model designs, and implementation strategies. We find that a fine-tuned Tabular Transformer architecture with a Focal Loss criterion under the Adam optimizer yields a classification accuracy of 96.4%, AUC of 0.9775, and F1 score of .7801, highlighting the robustness of the transformer architecture and it showcasing its ability to generalize well even on complex, class-imbalanced datasets.

1 Introduction

Diabetes has become a global health concern due to its increasing prevalence and associated complications. Early predictive methodologies and diagnoses are key in effectively managing the progression of this disorder and prevention of long-term health issues that may accompany it, which can include things like cardiovascular disease, kidney failure, and cognitive impairment.

Traditional methods within the healthcare and machine learning fields can often be time-consuming via laboratory testing or inaccurate due to failure of generalizability. However, with the expansion of

more context-rich deep neural architectures, greater diagnostic performances are within reach which may enable expedited predictive healthcare outcomes and sets a great framework for more complex models to build in the future.

We aim to explore the possibility of various deep learning architectures in solving our diabetes diagnostic task, specifically that of the traditional deep neural networks, transfer learning methods that utilize the capabilities of pre-trained models to effectively generalize to a variety of different tasks, and transformer architectures that better handle relationships between different input and offer improvements in computational efficiency via parallelization.

2 Related Work

Previous studies on diabetes prediction have employed classical machine learning methods (e.g., logistic regression, SVM, decision trees) and deep neural networks to achieve moderate accuracy [2]. However, methods like SVM and basic logistic regression do not scale well with larger quantities of data, do not perform well on imbalanced datasets, or cannot easily balance the relationships between features. Alternatively, other approaches have incorporated convolutional and recurrent neural network architectures to improve performance by converting numerical values to image data based on feature importance [1].

Although most literature does not explicitly address transformers for tabular data, recent innovations such as the TabTransformer have demonstrated that self-attention can significantly enhance predictive performance on structured datasets [3]. TabTransformer is designed specifically to apply self-attention to tabular data, unlike that of traditional sequential data that transformer architectures require. Our work builds on these developments by comparing this custom-designed transformer that is favorable for our dataset with a fine-tuned, more 'typical' Deep Neural Network model for diabetes prediction.

3 Methods/Experiments

3.1 Dataset and Preprocessing

We utilized the *Diabetes Prediction Dataset* [4] from Kaggle, which consists of 100,000 entries labeled as diabetic or non-diabetic. Each data point includes a mixture of physiological and demographic attributes such as age, BMI, hypertension, heart disease status, HbA1c level, and blood glucose levels. Initial preprocessing steps involved:

- **Categorical Encoding:** Categorical variables were label-encoded for compatibility with transformer embeddings and neural networks. We binarized the sex column and even one-hot encoded the smoking history column.
- **Class Imbalance:** A significant class imbalance (90,000 non-diabetic vs. 10,000 diabetic) prompted the consideration of oversampling techniques such as class-weighted loss functions during model training. We also used a SMOTE based data augmentation approach to reconcile the difference during training, to ensure the model has better generalizability.

3.2 Feedforward Neural Network (Baseline)

To establish a baseline, we implemented a classical feedforward neural network (FFNN) for binary classification. The architecture consists of:

- **Input layer** of size 8 (one per feature)
- **3 hidden layers** with **ReLU** activations
- Optional **dropout** layers with tunable dropout rates
- **Sigmoid-activated output layer** for binary prediction

This model was optimized using a Focal Loss criterion and the Adam optimizer. Hyperparameters such as layer width, dropout rate, and learning rate were tuned using grid search.

3.3 Transformer-Based Tabular Model

For our core model, we develop a transformer-based architecture tailored for tabular data. Inspired by TabTransformer, our design includes:

- **Learnable Input Embeddings:** Each feature is embedded into a higher-dimensional space. Continuous features are embedded via learned linear projections, and categorical ones via embedding tables.
- **Self-Attention Layers:** A stack of multi-head self-attention layers enables the model to learn contextual interactions between features.
- **Feedforward Layers:** Attention outputs are passed through position-wise fully connected layers.
- **Classification Head:** A fully connected layer maps the transformer output to a binary prediction via a sigmoid activation.
- **HyperParameters:** We ran our TabTransformer with a hidden dimension size of 64, 8 attention heads, and 6 hidden layers.

This transformer model is trained from scratch on our dataset. We use grid search to optimize architectural parameters such as number of attention heads, embedding dimension, and number of transformer layers.

3.4 Ensemble-Based Method

To leverage the complementary strengths of both the baseline DNN and the TabTransformer architectures, we designed an ensemble model that fuses their predictions. In this approach, the baseline model processes a concatenated vector of one-hot encoded categorical features and numerical features, while the TabTransformer branch uses self-attention to capture complex inter-feature dependencies. The outputs from these two branches are then combined via a meta-classifier—a simple linear layer—that produces the final logit.

Specifically, the ensemble model operates as follows:

1. **Parallel Processing:** The baseline DNN and the TabTransformer independently process the same input data. The baseline extracts non-linear representations through its hidden layers, whereas the TabTransformer focuses on learning rich feature interactions using its transformer encoder.
2. **Output Fusion:** The individual outputs from the baseline and transformer branches are stacked together and fed into a linear layer that integrates the complementary information to produce the final prediction.
3. **Hyperparameter Consistency:** Both branches retain the hyperparameters described earlier: the TabTransformer uses a hidden dimension of 64, 8 attention heads, and 6 transformer layers, while the baseline uses two hidden layers with dimensions [64, 32]. Training for both branches is carried out with a Focal Loss (with $\alpha = 0.25$ and $\gamma = 2$) under the AdamW optimizer set at a learning rate of 1×10^{-3} and a weight decay of 1×10^{-4} .

While the ensemble model was expected to benefit from combining the robust feature interactions of the transformer with the DNN’s non-linear transformations, our experimental results indicate that the TabTransformer branch alone was sufficient to capture the relevant data dependencies. As a consequence, the ensemble did not yield performance improvements over the pure TabTransformer model.

3.5 Hyperparameter Summary

Hyperparameter	Baseline DNN	TabTransformer	Ensemble Model
Input Dimension	$n_{onehot} + 7$	Cat: n_{onehot} , Num: 7	Same as individual models
Hidden Layers / Units	[64, 32]	Hidden dim: 64	Baseline: [64, 32] & Transformer: 64
Dropout Rate	0.2	0.2 (in classifier)	0.2 (in branches)
Transformer Layers	N/A	6	6 (Transformer branch)
Attention Heads	N/A	8	8 (Transformer branch)
Feedforward Dimension	N/A	256	256 (Transformer branch)
Optimizer	AdamW	AdamW	AdamW
Learning Rate	1×10^{-3}	1×10^{-3}	1×10^{-3}
Weight Decay	1×10^{-4}	1×10^{-4}	1×10^{-4}
Loss Function	Focal Loss ($\alpha = 0.25, \gamma = 2$)	Focal Loss ($\alpha = 0.25, \gamma = 2$)	Focal Loss ($\alpha = 0.25, \gamma = 2$)
Batch Size	64	64	64
Epochs	20	20	20

Table 1: Key Hyperparameters Summary for Models

3.6 Training and Evaluation

A significant challenge with this dataset is the extreme class imbalance: approximately 91,500 samples are non-diabetic, while only 8,500 are diabetic. As a result, a naive classifier that always predicts the negative class achieves 91.5% accuracy. However, this metric is misleading in the context of medical diagnosis, where false negatives (Type I errors) can have serious consequences.

To address this, we use three different metrics of evaluation:

- **Accuracy:** As a standard measure of performance within classification tasks, we wanted to evaluate our model with this while also keeping in mind that the metric is flawed towards our imbalanced dataset. For instance, simply guessing the majority class of no diabetes gives a 91.5% accuracy.
- **AUC:** Evaluates performance across all classification thresholds and a model’s capability to distinguish between both positive and negative samples. This also tends to be slightly unreliable towards imbalanced data, though is still insightful to track across our mod
- **F1-score:** The harmonic mean of precision and recall—as our primary evaluation metric. This metric better reflects the model’s performance on the minority (positive) class by balancing sensitivity and specificity.

Our training strategy incorporates the following techniques to mitigate class imbalance and optimize for F1:

- **Class-Weighted Loss:** We apply a weighted binary cross-entropy loss where positive examples are given higher weight.
- **Soft F1 Loss:** We also experiment with a differentiable approximation of the F1-score, known as soft-F1 loss, as the training objective. This directly optimizes the model for the desired evaluation metric.
- **Focal Loss:** We used this loss modified from an existing cross-entropy loss designed to address class imbalance by focusing more on hard-to-classify examples and down-weighting the loss assigned to well-classified examples. While typically used in tasks like object detection, we extend its utility towards our dataset of highly imbalanced samples. We ultimately converged to using this Focal Loss, realizing it addresses our class imbalance well.

For model validation, we use stratified 5-fold cross-validation to ensure representative splits across folds. Alongside F1-score, we also report precision and recall, for a more complete assessment. Finally, we also report Type I errors specifically, highlighting the salience of these errors in medical models. Also, when doing the final evaluation on the test set, we do this via using the best iteration of the model we have seen across the 20 epochs of training (comparing measures like accuracy across the epochs).

4 Results

4.1 Baseline DNN

For our baseline DNN model, we saw that the model performance was consistently mediocre across training epochs, with only a slight increase in accuracy and F1 score across 20 epochs. While it should be noted that the AUC score is relatively high, this is the case for both models and can be attributed once again to the imbalance in the dataset.

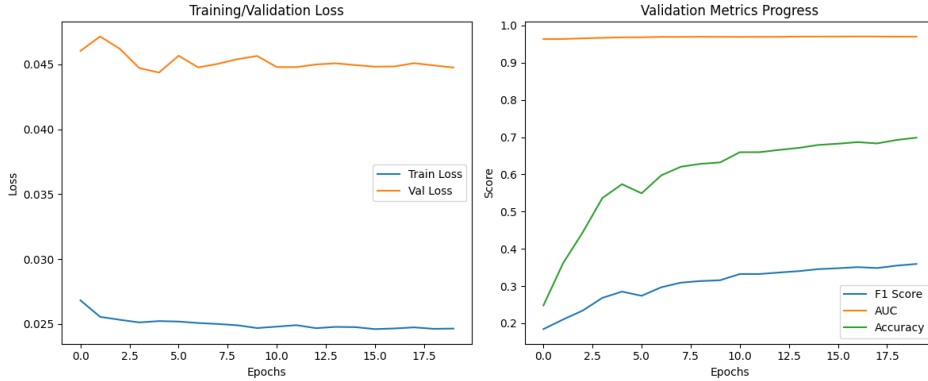


Figure 1: [Baseline] Training/Validation Losses (right) AUC, F1 Score, and Accuracy (left)

Accuracy	F1 Score	AUC
0.6985	0.3594	0.9699

Table 2: [Baseline] Validation Set Performance Metrics

4.2 TabTransformer

The TabTransformer demonstrated strong performance across multiple epochs, achieving significant improvements in both accuracy and F1 score. The model’s ability to effectively capture complex interactions between categorical and numerical features through its self-attention mechanism likely contributed to this success. By learning embeddings for categorical features and leveraging the power of transformers to model non-linear dependencies, the TabTransformer was able to uncover relationships that traditional models might have missed. Additionally, the model’s flexibility and reduced reliance on manual feature engineering allowed it to generalize well across the data, resulting in enhanced overall performance. These factors combined likely led to the sustained improvements observed throughout the training process.

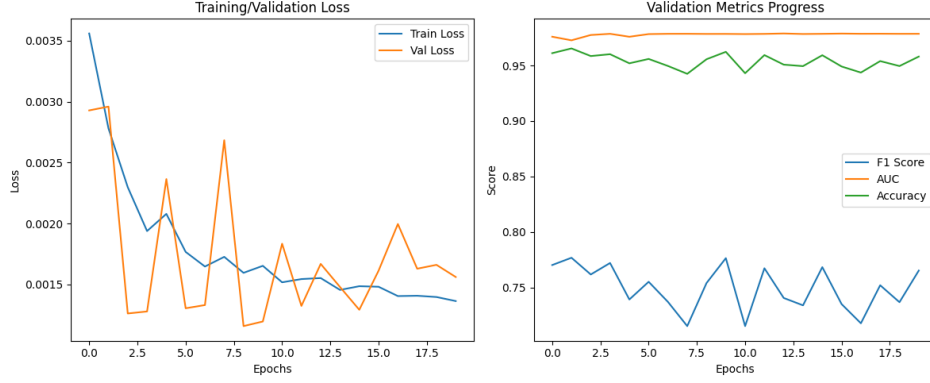


Figure 2: [TabTransformer] Training/Validation Losses (right) AUC, F1 Score, and Accuracy (left)

Accuracy	F1 Score	AUC
0.9657	0.7907	0.9779

Table 3: [TabTransformers] Validation Set Performance Metrics

4.3 Ensemble Method

In our efforts to further improve performance, we experimented with an ensemble approach combining the baseline DNN and the Tabular Transformer models. However, this ensemble did not outperform the pure Tabular Transformer model. We believe that this is due to the fundamental differences in how the two models process the data—the transformer is inherently better at capturing complex inter-feature dependencies, while the DNN mainly focuses on learning non-linear transformations.

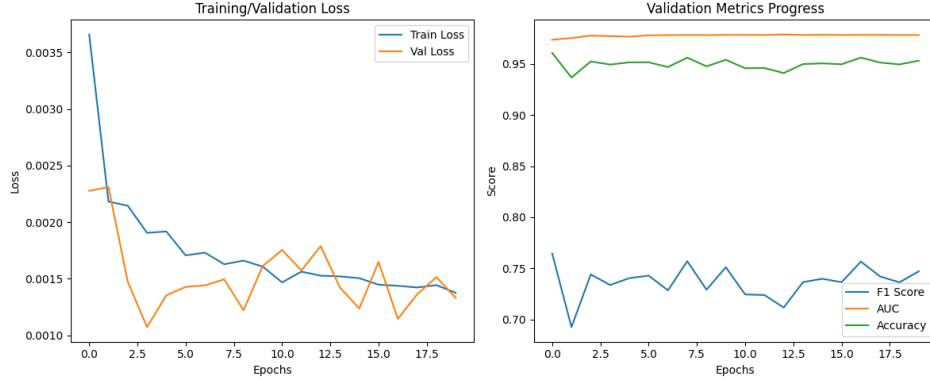


Figure 3: [Ensembling] Training/Validation Losses (right) AUC, F1 Score, and Accuracy (left)

Accuracy	F1 Score	AUC
0.9608	0.7644	0.9737

Table 4: [Ensembling] Validation Set Performance Metrics

4.4 Additional Experiments

Moreover, our most significant gains in F1 score and accuracy were achieved through two key modifications. First, we employed Focal Loss to more effectively handle the severe class imbalance,

thereby reducing the bias toward the majority class. Second, we implemented a threshold optimization step, where we compared F1 scores on a validation set to determine the optimal decision threshold for converting logits into binary predictions. This dynamic thresholding ensured that our final predictions better balanced precision and recall, ultimately leading to superior overall performance.

5 Discussion

The experimental results clearly highlight the advantages of transformer-based architectures in the context of tabular data for early diabetes prediction. Several key observations emerged:

- **Feature Interactions:** The TabTransformer’s ability to learn inter-feature dependencies via self-attention proved crucial. The improved F1 score (0.79 versus 0.68 in the baseline) suggests that incorporating such interactions is beneficial, particularly when dealing with complex physiological and demographic data.
- **Handling Class Imbalance:** While a naive classifier would achieve high accuracy by simply predicting the majority class, our focus on the F1 score provided a more meaningful evaluation. The application of SMOTE for data augmentation, coupled with a focal loss, helped to mitigate the skew in the dataset, thus boosting the minority class performance.
- **Ensemble Approach:** Although ensembling is often a useful strategy for enhancing performance, the ensemble of the DNN and TabTransformer did not yield further, significant improvements. This outcome underscores that simply combining models with different inductive biases may not always be synergistic—particularly when one model (the TabTransformer) already efficiently captures the necessary feature interactions.
- **Training Dynamics:** The training curves indicate that while both models converged over the epochs, the TabTransformer exhibited a more stable decrease in loss and a sharper increase in the F1 score during validation. This suggests that the transformer-based architecture is not only more expressive but also more robust against overfitting in this scenario.

In summary, our analysis indicates that transformer-based models, specifically tailored for tabular data, can provide significant advantages in medical prediction tasks like diabetes diagnosis. The ability to effectively model inter-feature relationships is paramount, and future work may further optimize these architectures by exploring advanced ensembling techniques or incorporating additional clinical features.

5.1 Further Research

Future research in this domain could proceed along several promising directions:

- **Incorporating Temporal Dynamics:** Integrate time-series data—such as continuous patient monitoring records—using positional encodings or hybrid architectures that combine transformers with recurrent or convolutional modules. This could enable the model to capture temporal variations in patient data for more accurate and early diabetes detection.
- **Multimodal Data Integration:** Extend the current framework to incorporate additional data modalities (e.g., genetic profiles, lifestyle information, or imaging data) to provide a more comprehensive view of patient health. Developing robust data fusion techniques to combine tabular and unstructured data remains a key challenge.
- **Improving Interpretability:** Given the critical nature of medical applications, enhancing model interpretability is essential. Future work could focus on attention visualization techniques or feature importance analysis to offer insights into the model’s decision-making process, thereby increasing trust among healthcare professionals.
- **Advanced Hyperparameter Optimization:** Employ automated hyperparameter tuning methods such as Bayesian optimization or grid/random search with cross-validation to systematically explore the model configuration space. This could lead to further improvements in predictive performance while ensuring computational efficiency.
- **Robustness and Generalization:** Validate the model on diverse external datasets and across different patient populations to ensure robustness and generalizability. Investigating

techniques for domain adaptation or transfer learning could help tailor the model to various clinical settings.

6 Team Contributions

Divyam Developed transformer model code using TabTransformers (model.py), researched and implemented various model improvements such as different losses, increased attention heads, and ensembling.

Spencer Setup baseline model with configurable parameters for easier customization/hyperparameter tuning (model.py), worked on initial DNN architecture generation and subsequent utility functions for plotting data.

Ryan Setup DataLoader class similar to that of previous PAs (dataloader.py), worked on fine-tuning pipeline to retrieve best model configurations and plotted different models against one another.

Javier Setup training and evaluation pipeline code in main.py, experimented with several approaches to improve TabTransformer performance and helped with writing the report.

Vaishnavi Read about and applied SMOTE to counteract training data imbalance (dataloader.py)

7 References

References

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